

4-1 Quadratic Functions and Transformations

Algebra 2 · Chapter 4: Quadratic Functions and Equations

Name: _____

Date: _____

Period: _____

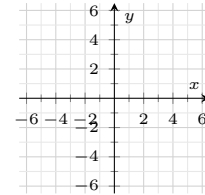
I can: recognize the parent function $y = x^2$ · describe how a , h , k transform the graph · find the vertex, axis, max/min, domain and range from vertex form · write a quadratic function from a graph or a description.

Vocabulary: parabola, vertex, axis of symmetry, vertex form, minimum/maximum value, reflection, translation, vertical stretch, vertical compression

1. Warm-Up: The Parent Function $f(x) = x^2$

Complete the table, then plot the points.

x	-3	-2	-1	0	1	2	3
$f(x)$							



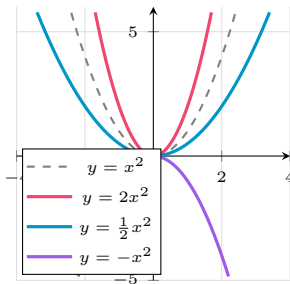
Notice

The outputs for x and $-x$ are _____, so the graph is _____.

Definition

A **quadratic function** can be written as $f(x) = ax^2 + bx + c$, where _____. Its graph is a U-shaped curve called a _____. The **vertex** is the _____; the **axis of symmetry** is the _____ through the vertex. For $f(x) = x^2$: vertex _____, axis _____.

2. What Does a Do? $f(x) = ax^2$



Notice

- $y = 2x^2$ is _____ than $y = x^2$ (each y is _____).
- $y = \frac{1}{2}x^2$ is _____ than $y = x^2$.
- $y = -x^2$ opens _____; a _____ in the x -axis.
- The vertex stays at _____.

Value of a	Effect on $y = x^2$
$ a > 1$	vertical _____ (narrower)
$0 < a < 1$	vertical _____ (wider)
$a > 0$	opens _____; vertex is a _____
$a < 0$	opens _____; vertex is a _____

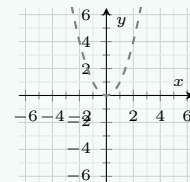
Common Mistake

Stretch/compression depends on $|a|$, not the sign: $y = -3x^2$ is a reflection *and* a vertical _____.

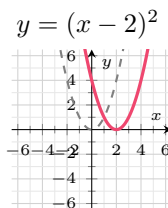
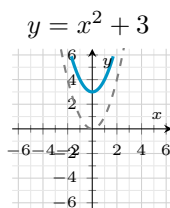
Watch Me 1 — Graph $f(x) = -\frac{1}{2}x^2$

x	-4	-2	0	2	4
$f(x)$					

Compared with $y = x^2$: reflection in the _____; vertical _____ by a factor of _____; vertex _____ is a _____.



3. What Do h and k Do?



Notice

$y = x^2 + 3$ moves the graph _____.

$y = (x - 2)^2$ moves the graph _____.

$x - 2$ inside the square moves the graph the _____ way of the sign.

Function	Translation of $y = x^2$	Vertex
$y = x^2 + k$	if $k > 0$, _____ if $k < 0$	_____
$y = (x - h)^2$	if $h > 0$, _____ if $h < 0$	_____
$y = (x + 3)^2$	_____ (here $h =$ _____)	_____

Common Mistake

$(x + 3)^2 = (x - (-3))^2$, so $h = -3$: the graph moves **left**, not right.

4. Vertex Form

Vertex Form $f(x) = a(x - h)^2 + k$

- vertex: _____ axis: _____
- $a > 0$: opens up, min value _____
- $a < 0$: opens down, max value _____
- domain: _____
- range: _____ if $a > 0$; _____ if $a < 0$
- $|a|$ controls _____

Order to graph: _____ /stretch with $a \rightarrow$ shift _____ by $h \rightarrow$ shift _____ by k

Watch Me 2 — $f(x) = (x + 3)^2 - 4$

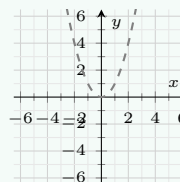
Rewrite: $f(x) = 1(x - (\quad))^2 + (\quad)$

$a =$ _____, $h =$ _____, $k =$ _____

Transformations: _____ 3, _____ 4

Vertex _____ Axis _____ value _____

Domain: _____ Range: _____



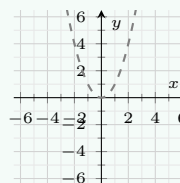
Watch Me 3 — $f(x) = -2(x - 1)^2 + 5$

$a =$ _____, $h =$ _____, $k =$ _____

Transformations (in order): _____, vertical _____ by _____,

_____ 1, _____ 5

Vertex _____ Axis _____ value _____ Range: _____

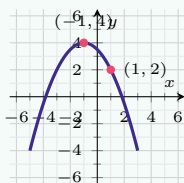


Your Turn — Read the Vertex Form

Function	Vertex	Axis	Max or Min?	Value
$f(x) = (x - 5)^2 + 2$	_____	_____	_____	_____
$g(x) = -(x + 1)^2 - 3$	_____	_____	_____	_____
$p(x) = \frac{1}{2}x^2 - 6$	_____	_____	_____	_____

5. Writing Equations

Watch Me 4 — From a Graph



Step 1. Vertex $(h, k) = \underline{\hspace{2cm}}$, so $f(x) = a(x + \underline{\hspace{1cm}})^2 + \underline{\hspace{1cm}}$

Step 2. Use the point $(1, 2)$: $2 = a(1 + 1)^2 + 4 \Rightarrow 2 = \underline{\hspace{2cm}} \Rightarrow a = \underline{\hspace{2cm}}$

Step 3. $f(x) = \underline{\hspace{4cm}}$

Check: it opens $\underline{\hspace{1cm}}$, so a should be $\underline{\hspace{1cm}}$ ✓

Watch Me 5 — From a Description

Start with $y = x^2$. Reflect it in the x -axis, stretch it vertically by 3, then translate it **4 units left** and **1 unit down**.

reflect + stretch by 3 $\Rightarrow a = \underline{\hspace{2cm}}$ 4 left $\Rightarrow h = \underline{\hspace{2cm}}$ 1 down $\Rightarrow k = \underline{\hspace{2cm}}$ $g(x) = \underline{\hspace{4cm}}$

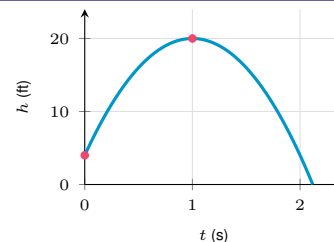
What If...?

1. a changes from 2 to -2 . Does the vertex move? $\underline{\hspace{4cm}}$
2. Swap h and k : $(x - 2)^2 + 5$ vs. $(x - 5)^2 + 2$? $\underline{\hspace{4cm}}$
3. $a = 0$: is $y = 0(x - 1)^2 + 3$ a parabola? $\underline{\hspace{4cm}}$

6. Connection: A Water Fountain

The height (feet) of a water stream t seconds after it leaves the fountain is $h(t) = -16(t - 1)^2 + 20$.

1. Maximum height: $\underline{\hspace{2cm}}$ It reaches the top after $\underline{\hspace{2cm}}$.
2. Starting height $h(0) = \underline{\hspace{2cm}}$
3. Why is the vertex a maximum? $\underline{\hspace{4cm}}$



7. Practice

Practice

1. Describe the transformations of $y = x^2$: $g(x) = 3(x - 2)^2 + 1$
 $\underline{\hspace{4cm}}$
2. Describe the transformations of $y = x^2$: $g(x) = -\frac{1}{4}(x + 5)^2$
 $\underline{\hspace{4cm}}$
3. Write the function: $y = x^2$ reflected in the x -axis, then 6 right and 2 up. $g(x) = \underline{\hspace{4cm}}$
4. Vertex $(3, 1)$; the graph passes through $(5, 9)$. $f(x) = \underline{\hspace{4cm}}$
5. **Challenge.** $f(x) = a(x - h)^2 + k$ passes through $(0, 5)$ and $(4, 5)$, and its minimum value is -3 . Find $f(x)$.

$f(x) = \underline{\hspace{4cm}}$

8. Summary & Exit Ticket

Summary

Parent function $y = x^2$, vertex _____, axis _____
Vertex form $f(x) = a(x - h)^2 + k$, vertex _____
 a sign: opens up/down; $|a|$: _____
 h horizontal shift — the _____ of the sign in $(x - h)$
 k vertical shift; also the _____ value

Exit Ticket

For $f(x) = -3(x + 2)^2 + 7$: vertex _____, axis _____, opens _____, _____ value _____, range _____.
Transformations of $y = x^2$: _____

More practice: fangmath.com → Algebra 2 → Chapter 4 → Interactive practice (Quadratic Transformations).